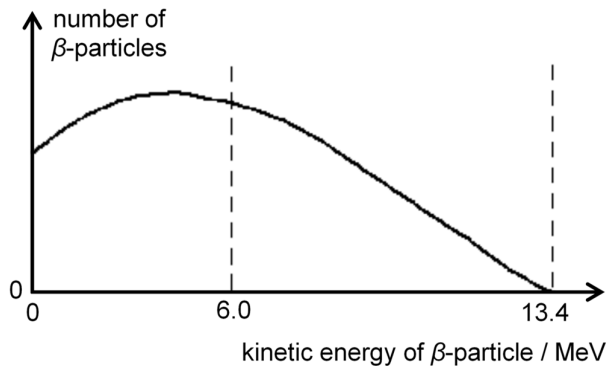


28 The beta spectrum for ^{12}B decay is as shown below.



The kinetic energy of an emitted β -particle is 6.0 MeV. What is the approximate energy of the associated neutrino?

A 4.0 MeV

B 6.0 MeV

C 7.4 MeV

D 13.4 MeV

Based on COE and COM, the kinetic energy of the daughter nuclei is negligible hence the total energy released is shared between β particle and neutrino. Since the highest possible KE of β particle is 13.4 MeV, i.e. when neutrino has zero KE, hence the total energy released by the reaction is also 13.4 MeV, and this value is fixed for this particular reaction. Thus, when an emitted β particle has KE of 6.0 MeV, the associated neutrino must have $13.4 - 6.0 = 7.4$ MeV of energy.

29 Two radioactive isotopes P and Q have half-lives of 10 minutes and 15 minutes respectively. A